# Regular Expressions

## Chapter 11

Python for Everybody
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# Regular Expressions

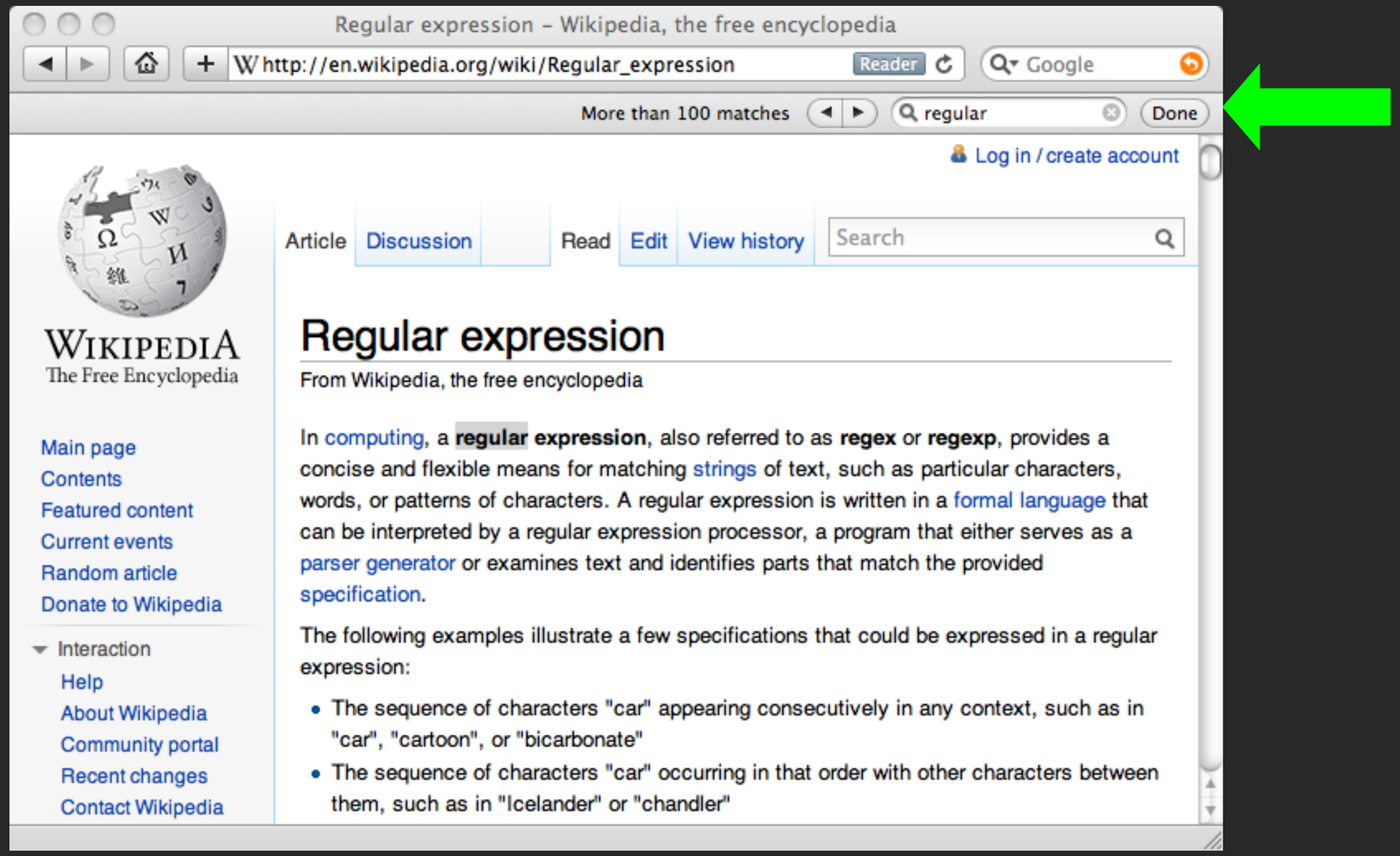
In computing, a regular expression, also referred to as “regex” or “regexp”, provides a concise and flexible means for matching strings of text, such as particular characters, words, or patterns of characters. A regular expression is written in a formal language that can be interpreted by a regular expression processor.

http://en.wikipedia.org/wiki/Regular_expression

# Regular Expressions

Really clever “wild card” expressions for matching and parsing strings

http://en.wikipedia.org/wiki/Regular_expression




Really smart “Find” or “Search”

# Understanding Regular Expressions

- Very powerful and quite cryptic
- Fun once you understand them
- Regular expressions are a language unto themselves
- A language of “marker characters” - programming with characters
- It is kind of an “old school” language - compact

http://xkcd.com/208/

# Regular Expression Quick Guide

```
^         Matches the beginning of a line
$         Matches the end of the line
.         Matches any character
\s        Matches whitespace
\S        Matches any non-whitespace character
*         Repeats a character zero or more times
*?        Repeats a character zero or more times (non-greedy)
+         Repeats a character one or more times
+?        Repeats a character one or more times (non-greedy)
[aeiou]   Matches a single character in the listed set
[^XYZ]    Matches a single character not in the listed set
[a-z0-9]  The set of characters can include a range
(         Indicates where string extraction is to start
)         Indicates where string extraction is to end
```

https://www.py4e.com/lectures3/Pythonlearn-11-Regex-Handout.txt

# The Regular Expression Module

- Before you can use regular expressions in your program, you must import the library using "import re"
- You can use re.search() to see if a string matches a regular expression, similar to using the find() method for strings
- You can use re.findall() to extract portions of a string that match your regular expression, similar to a combination of find() and slicing: var[5:10]

# Using re.search() Like find()

```
hand = open('mbox-short.txt')
for line in hand:
    line = line.rstrip()
    if line.find('From:') >= 0:
        print(line)
```

```
import re

hand = open('mbox-short.txt')
for line in hand:
    line = line.rstrip()
    if re.search('From:', line) :
        print(line)
```

# Using re.search() Like startswith()

```
hand = open('mbox-short.txt')
for line in hand:
    line = line.rstrip()
    if line.find('From:') >= 0:
        print(line)
```

```
import re

hand = open('mbox-short.txt')
for line in hand:
    line = line.rstrip()
    if re.search('From:', line) :
        print(line)
```

We fine-tune what is matched by adding special characters to the string

# Wild-Card Characters

- The dot character matches any character
- If you add the asterisk character, the character is “any number of times”

```
X-Sieve: CMU Sieve 2.3
X-DSPAM-Result: Innocent
X-DSPAM-Confidence: 0.8475
X-Content-Type-Message-Body: text/plain
```

```
^X.*:
```

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```

Match the start of the line

Many times

```
^X.*:
```

Match any character

# Fine-Tuning Your Match

Depending on how “clean” your data is and the purpose of your application, you may want to narrow your match down a bit

```
X-Sieve: CMU Sieve 2.3
X-DSPAM-Result: Innocent
X-Plane is behind schedule: two weeks
```

Match the start of the line

Many times

^X.*:

Match any character

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```
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X-Plane is behind schedule: two weeks
```

Match the start of the line

One or more times

```
^X-\S+:
```

Match any non-whitespace character